

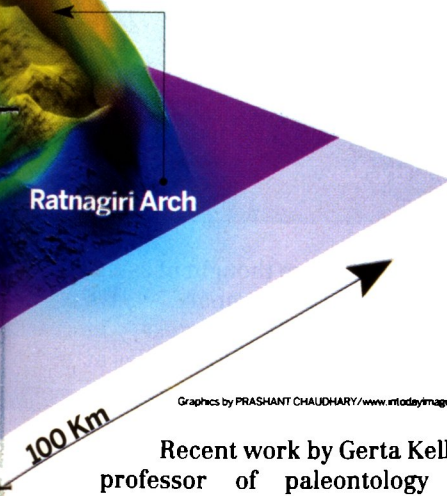
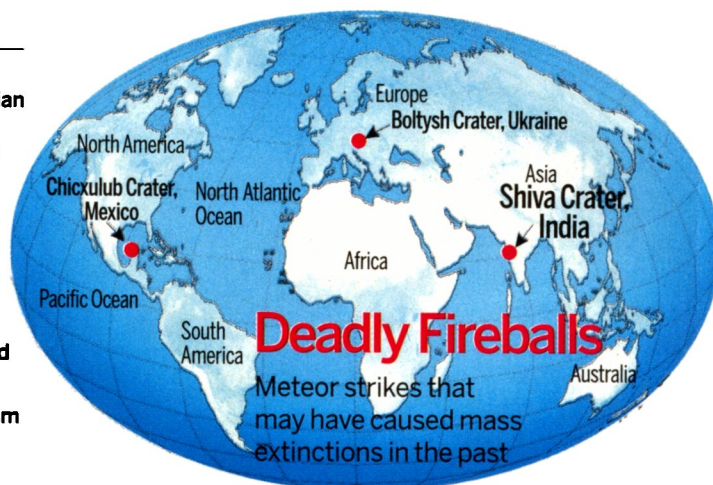
The Big Killers



SHIVA CRATER This meteorite crash in the Arabian Sea, 65 million years ago, is now theorised to have been the primary cause of the extinction of dinosaurs.

CHICXULUB CRATER Located off the coast of Mexico's Yucatan, most experts considered this meteor crash to have caused the death of the dinosaurs.

THE OTHER VIEW Volcanism was the main cause of the dinosaurs' downfall.



Graphics by PRASHANT CHAUDHARY/www.ertodayimages.com

Recent work by Gerta Keller, professor of paleontology at Princeton University, has suggested that the Chicxulub event took place 3,00,000 years earlier than the K-T boundary—a distinctive geological signature which dates to the K-T extinction event—when dinosaurs died out. Thus the Chicxulub impact cannot be the proximate cause of the dinosaur extinction. She advocates that volcanism is the main cause for the dinosaurs' downfall and downplays Chatterjee's claims. But her idea has long been controversial and remains that way.

"When Deccan lava was erupting, dinosaurs in India were still thriving. We have found dinosaur bones and their eggs preserved in the sedimentary layer between Deccan lava flows in many areas in India. Thus the Deccan pollutions might have been an accomplice for the environmental

crisis but not the direct cause of the extinction. Indirectly, Deccan eruptions might have been responsible for the acidification of the oceans that led to food collapse of the marine organisms. But for the dinosaur extinction, the Shiva impact seems to be the proximate cause," argues Chatterjee, who presented his holistic hypothesis to the Geological Society of America at its recent annual meeting in Portland, Oregon. By researching the Shiva Crater, scientists hope to learn more about the life of the Earth and the important events in the planet's history. If the crater was indeed created by a catastrophic collision, it could help explain why so many species went extinct during the K-T event, and it might also elucidate upon the mineral composition of the K-T boundary associated with this period in geologic history.

Based on newer evidence in his decade-old study, Chatterjee says, "The Shiva Crater showed that the meteorite struck at the same time as the mass extinction and created enough catastrophic force to destroy 70 per cent of Earth's plant and animal communities on the land as well as in the seas." In effect, the Chicxulub crater is dismissed as the creation of a milder sledgehammer-like blow, the teaser before the big one. Chatterjee is relying on both geophysical evidence and core samples collected by the Oil and Natural Gas

Corporation about geology and geophysics of the Bombay High region. He finds this information quite valuable to reconstruct the geometry and determine the age of the Shiva Crater, a massive sea floor structure that is tear-drop-shaped and roughly 600 by 400 km in area. Although craters of this size are common on the surface of the Moon, before the Shiva, the largest impact crater was Vredefort in South Africa with a diameter of 300 km. The diameter of this Arabian Sea crater is about 500 km.

Though the actual crater is covered by sediment, Chatterjee says the geological evidence he collected allowed him to map it out. It is hard to imagine such a cataclysm. If his postulate is right, the Shiva impact vapourised the Earth's crust at the point of collision, leaving nothing but ultra-hot mantle material to well up in its place. The impact appears to have sheared or destroyed much of the 30-mile-thick granite layer off the western coast of India. It is likely that the strike triggered and enhanced the main eruptive phase of the massive continental flood basalts, such as the Deccan Traps and related volcanic eruptions that cover much of Western India. This activity resulted in vast quantities of lava flooding the western part of India in a short amount of time. The greatest lava event coincides with the meteorite crash.

What's more, the impact broke the